How Literacy and Age at Marriage Shape Family Size in Portugal, 1979*

A Statistic Research Using Poisson (Negative Binominal) Regression Models

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1 Introduction

Family size is not just the result of economic factors. On a deeper level, it reflects a person's upbringing, social and cultural norms, and the uneven development of different regions. Most past studies have focused on how education affects women's fertility, but they have paid less attention to the role of marriage age and regional differences. In a country like Portugal in

*Code and data are available at: https://github.com/Jie-jiao05/Family_Size.

1979, which was going through social changes, the interaction between education and marriage age could have had a more complex impact on fertility patterns across different regions.

Previous studies provide important background on how education, marriage age, and regional differences influence fertility decisions. (Teresa Castro Martin 1995) examined fertility patterns in 26 countries and found a negative correlation between women's education levels and fertility rates. This was not only because literacy helped women access contraceptive knowledge but also because education changed how they planned their futures, making it a key variable for study. (Naila Kabeer 1996) highlighted that delaying marriage generally led to fewer children by shortening the reproductive window and allowing women more time for career development. However, the reasons behind later marriage are complex—do women choose to marry later because they have more autonomy, or is it a response to economic conditions? This makes marriage age another important variable to consider. (Conceição Rego, Conceição Freire, Isabel Joaquina Ramos, Andreia Dionísio, Maria da Saudade Baltazar, Maria Raquel Lucas 2016) studied regional disparities in Portugal and found that urbanization had a major impact on family structure. Cities provided better education and job opportunities, influencing women's fertility decisions. Their findings suggest that when analyzing fertility choices in Portugal, it is not enough to look at individual factors like education and marriage age—regional differences must also be considered.

Were a woman's education level and age enough to influence her decisions about childbirth? Did fertility patterns differ significantly across regions? To answer these questions, we will use Generalized Linear Models (GLMs), including Poisson regression and Negative Binomial regression, to analyze how these factors affected family size. This study will not only help us understand Portugal's fertility transition but also provide new insights into how education, marriage age, and fertility rates are connected on a global scale.

2 Method

Before model fitting, we processed key categorical variables to ensure their compatibility with regression analysis. Literacy, region, and marriage age were converted into dummy variables, allowing us to include them as predictors in our models. Region was classified into five categories: Lisbon, Porto, areas with populations under 10k (lt10k), areas with populations between 10-20k (10-20k), and areas with populations over 20k (20k+). Marriage age was grouped into discrete categories to analyze its effect on family size. To verify the importance of these variables, we conducted an ANOVA test to assess whether literacy, region, and marriage age significantly influenced family size. Variables with p-values below 0.05 were retained in the model.

Since family size is a count variable, we employed Generalized Linear Models (GLMs), specifically Poisson regression and Negative Binomial regression. The Poisson model is a standard choice under the assumption that the mean and variance of the response variable are equal. To ensure an appropriate model fit, we first examine region histograms to identify potential sample size imbalances across different areas. If certain regions have disproportionately large sample sizes, this could contribute to overdispersion in the Poisson model. In such cases, a Negative Binomial model provides a better fit by incorporating an additional dispersion parameter to account for extra variability. We considere to include an offset because marriage duration directly affects the number of children. Without adjustment, individuals with longer marriages may have larger family sizes simply due to a longer reproductive window, leading to biased model estimates. We choose log(monthsSinceMarriage) as the offset because marriage duration is a cumulative variable, and taking the logarithm converts multiplicative relationships into additive ones, making the model more stable and easier to interpret. We then compared Poisson regression without an offset, Poisson regression with an offset, and Negative Binomial regression with an offset to evaluate which specification provided the best fit.

To assess model effectiveness, we visualize the Plot Coefficient Estimates graph, which shows the impact of model adjustments on coefficient stability. The final model is selected based on AIC, Log-Likelihood values, and p-values, ensuring the best model fit. We choose the model with the lowest AIC and highest Log-Likelihood as the final specification. Once the final model is determined, we generate a summary table displaying coefficient estimates along with their confidence intervals. Additionally, to improve interpretability, we convert the coefficients into rate ratios, providing a clearer understanding of the effects of literacy, marriage age, and region on family size.

3 Result

3.1 Data Description

This dataset originates from the 1979-80 Portuguese Fertility Survey (Instituto Nacional de Estatística (Portugal) 1979), conducted as part of the World Fertility Survey (WFS) program and archived by Princeton University's Office of Population Research (OPR).

The Dataset includes 5,148 observations and 8 variables, covering factors such as literacy, marriage age, pregnancies, children, and regional demographics. Table 1 it shows a statistical summary of our dataset.

To analyze the relationship between family size, literacy, and age at marriage, the variable Literacy was categorized as a dummy variable, assigning values 0 and 1 to illiterate and literate individuals, respectively. The analysis primarily focused on demographic factors, economic influences, and regional differences, aiming to assess their impact on family size. Additionally, an offset variable accounting for the duration of marriage was included to control for varying exposure times.

Table 1: Dataset Summary

age	ageMarried	monthsSinceM	pregnancies
Min. :15.00	0to15 : 52	Min. : 0.0	Min. : 0.000
1st Qu.:28.75	15to18 : 452	1st Qu.: 73.0	1st Qu.: 1.000
Median :36.00	18to20 : 910	Median :148.0	Median : 2.000
Mean :35.30	20to22 :1126	Mean :155.2	Mean : 2.282
3rd Qu.:43.00	22to25 :1468	3rd Qu.:230.0	3rd Qu.: 3.000
Max. :49.00	25to30 : 923	Max. :431.0	Max. :16.000
	30toInf: 217		

children	sons	region	literacy
Min. : 0.00	Min. : 0.000	10-20k: 433	no : 581
1st Qu.: 1.00	1st Qu.: 0.000	20k+ : 583	yes:4567
Median : 2.00	Median : 1.000	lisbon: 470	
Mean : 2.26	Mean : 1.187	lt10k :3502	
3rd Qu.: 3.00	3rd Qu.: 2.000	porto : 160	
Max. :17.00	Max. :10.000		

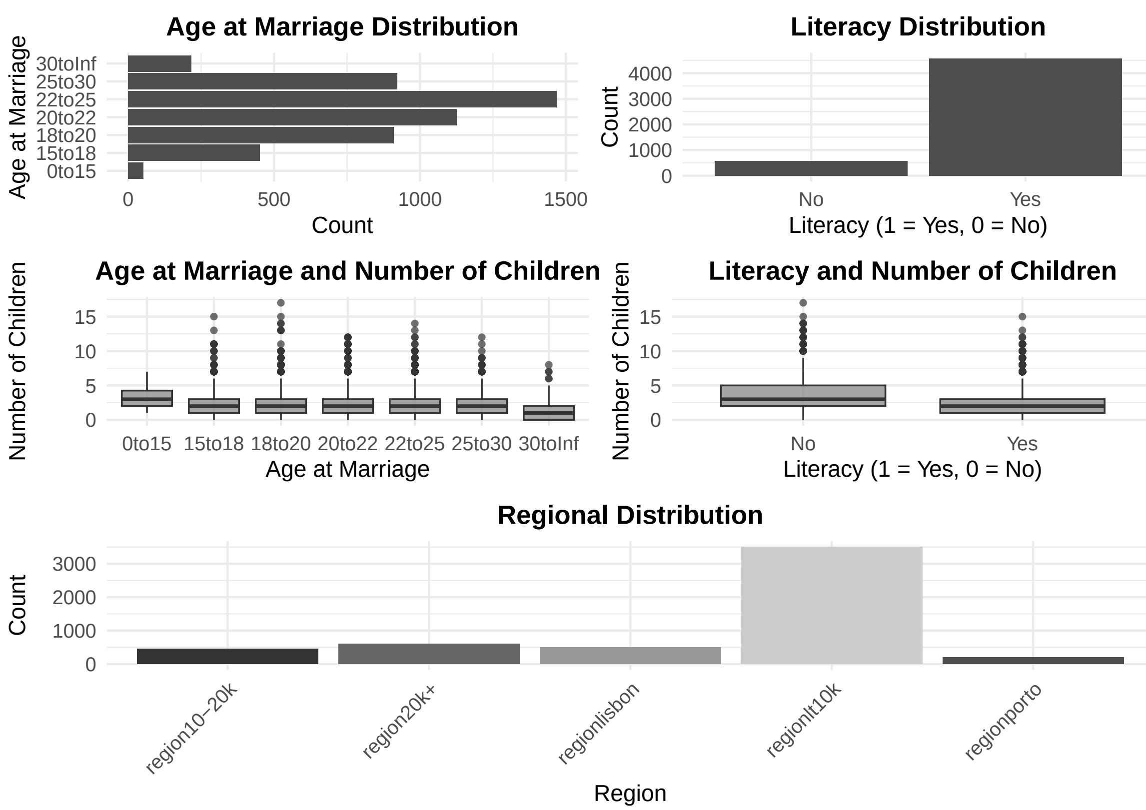

Figure 1: Distribution of Key Variable

Figure 1 illustrates the distribution of key variables and their relationship with family size. In general, the age of marriage is normally distributed within the dataset, slightly skewed to the right, with a peak at 22-25 years old. At the same time, the Portuguese have a low level of education, most of them come from the lt10k area, and illiterate people exist only as a minority in society. From the boxplot of the age of marriage and the number of children, we can see that the number of children is inversely proportional to the age. The lower the age of marriage, the more children there are. At the same time, those who are not educated generally have more children.

But unignorable, the vast majority of samples come from region_lt10k, creating a highly imbalanced data distribution, which may lead to increased variance and potential overdispersion. To account for differences in exposure across groups, we will consider an offset term during model selection, adjusting for variations in time since marriage. Details will be discussed in Section 3.2 and Section 3.2.2.

3.2 Model Selection

3.2.1 Anova Test of Region, Marry Age and Literacy

Our research aims to analyze how marriage age and literacy impact family size across different regions. To achieve this, we use age at marriage, literacy, and region as predictor variables, with children (family size) as the response variable.

To assess the significance of each predictor, we conducted an ANOVA test, and the results are shown in Table 2. The table indicates that all p-values are less than 0.05, allowing us to reject the null hypothesis. This confirms that literacy, marriage age, and region significantly influence family size.

3.2.2 Over-Dispersion and Offset

In the previous coefficient plot (Figure 1), the region distribution plot shows that the majority of samples come from region_lt10k, raising concerns about potential overdispersion. To address this issue, we explore two adjustments: adding an offset and transforming the model into a Negative Binomial distribution.

As shown in Figure 2 , incorporating an offset (log(monthsSinceM)) reduces the magnitude of estimates (from blue to orange), help to stabilize variance. And Transform into Negative Binomial distribution(green) further refines confidence intervals while preserving coefficient. And the dispersion ratio is 1.24, this suggests a mild overdispersion, making the Negative Binomial model with an offset the a better choice.

Therefore, in the following Section 3.2, we will structure our model selection process based on the presence or absence of an offset and whether the model follows a Negative Binomial distribution. This will allow us to evaluate whether incorporating these adjustments improves model performance.

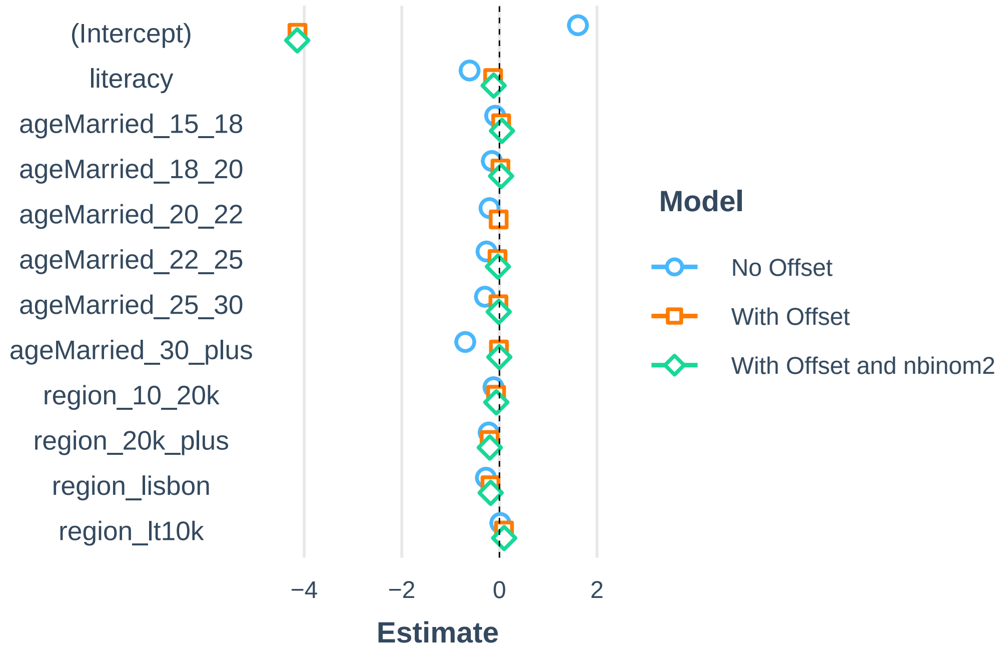

Figure 2: Distribution of Key Variable

3.2.3 Model Setting

For model Selection we consider 3 model under following condition and conduct Log_Likelihood test, AIC, Deviance to make comparison cross model, and comes with our final model. The detailed models are list below. And for the consideration of offset and over-dispersion are shown in Section 3.2.2

Model 1: Poisson Regression with Offset

$$\begin{aligned}
\text{children}_i &\sim \text{Poisson}(\lambda_i) && (1)\\
\log(\lambda_i) &= \beta_0 + \beta_1 \cdot \text{literacy}_i && (2)\\
&\quad + \beta_2 \cdot \text{ageMarried}_{15-18,i} + \beta_3 \cdot \text{ageMarried}_{18-20,i} && (3)\\
&\quad + \beta_4 \cdot \text{ageMarried}_{22-25,i} + \beta_5 \cdot \text{ageMarried}_{25-30,i} && (4)\\
&\quad + \beta_6 \cdot \text{ageMarried}_{30+,i} + \beta_7 \cdot \text{region}_{10-20k,i} && (5)\\
&\quad + \beta_8 \cdot \text{region}_{20k+,i} + \beta_9 \cdot \text{region}_{Lisbon,i} && (6)\\
&\quad + \beta_{10} \cdot \text{region}_{lt10k,i} + \log(\text{monthsSinceM}_i) && (7)
\end{aligned}$$

Model 2: Negative Binomial Regression with offset

$$
\begin{aligned}
\text{children}_i &\sim \text{Negative Binomial}(\lambda_i, \theta) && (8)\\
\log(\lambda_i) &= \beta_0 + \beta_1 \cdot \text{literacy}_i && (9)\\
&\quad + \beta_2 \cdot \text{ageMarried}_{15-18,i} + \beta_3 \cdot \text{ageMarried}_{18-20,i} && (10)\\
&\quad + \beta_4 \cdot \text{ageMarried}_{22-25,i} + \beta_5 \cdot \text{ageMarried}_{25-30,i} && (11)\\
&\quad + \beta_6 \cdot \text{ageMarried}_{30+,i} + \beta_7 \cdot \text{region}_{10-20k,i} && (12)\\
&\quad + \beta_8 \cdot \text{region}_{20k+,i} + \beta_9 \cdot \text{region}_{Lisbon,i} && (13)\\
&\quad + \beta_{10} \cdot \text{region}_{lt10k,i} + \log(\text{monthsSinceM}_i) && (14)
\end{aligned}
$$

Model 3: Poisson Regression without Offset

$$
\begin{aligned}
\text{children}_i &\sim \text{Poisson}(\lambda_i) && (15)\\
\log(\lambda_i) &= \beta_0 + \beta_1 \cdot \text{literacy}_i && (16)\\
&\quad + \beta_2 \cdot \text{ageMarried}_{15-18,i} + \beta_3 \cdot \text{ageMarried}_{18-20,i} && (17)\\
&\quad + \beta_4 \cdot \text{ageMarried}_{22-25,i} + \beta_5 \cdot \text{ageMarried}_{25-30,i} && (18)\\
&\quad + \beta_6 \cdot \text{ageMarried}_{30+,i} + \beta_7 \cdot \text{region}_{10-20k,i} && (19)\\
&\quad + \beta_8 \cdot \text{region}_{20k+,i} + \beta_9 \cdot \text{region}_{Lisbon,i} && (20)\\
&\quad + \beta_{10} \cdot \text{region}_{lt10k,i} && (21)
\end{aligned}
$$

Table 2: Likelihood Ratio Test Results

Comparison	Log_Likelihood	Chi_Square	p_Value	AIC
Model 1	-8944.7	NA	NA	17913.42
Model 2	-8909.1	NA	NA	17842.12
Model 3	-9234.8	NA	NA	18493.57
Model 1 vs Model 2	NA	71.307	2.2e-16	NA
Model 1 vs Model 3	NA	580.150	2.2e-16	NA
Model 2 vs Model 3	NA	651.450	2.2e-16	NA
Full Model vs No Literacy	NA	19.905	8.137e-06	NA
Full Model vs No Region	NA	118.890	2.2e-16	NA
Full Model vs No Marriage Age	NA	748.720	2.2e-16	NA

From Table 2, we observe that, based on the Log-Likelihood test, Model 2 provides the most accurate results as it effectively accounts for both overdispersion and the offset. Additionally, Model 2 has the lowest AIC (17,842.12) and the biggest Log-Likelihood(-8909.1) , indicating a better overall fit compared to Models 1 and 3. Therefore, we select Model 2 as our final model.

3.3 Final Model

Our final model, Model 2, accounts for over-dispersion with an offset, and its statistical summary is presented in Table 3. By analyzing the results, we observe that when RegionPorto and Agemarried_0_15 are set as the reference groups, the intercept, literacy, and regional factors (20k+ and Lisbon) are statistically significant.

Using marriage age 0–15 and Porto as reference categories, literate individuals have 11.2% fewer children than illiterate individuals, suggesting that lower education levels are associated with larger family sizes. Additionally, regional disparities are evident, with families in Lisbon having a family size that is 0.835 times that of those in Porto, indicating that less developed regions tend to have larger families. Compare to Lisbon and Porto, a literate individual in Lisbon married between 22–25 is expected to have 25.9% fewer children than an illiterate counterpart in Porto.

Comparing Lisbon and Porto, a literate individual in Lisbon married between 22–25 is expected to have 25.9% fewer children than an illiterate counterpart in Porto, reinforcing the combined impact of education, regional context, and marital timing on fertility outcomes.

Final Model: Model 2: Negative Binomial Regression with offset

$$\begin{aligned}
\text{children}_i &\sim \text{Negative Binomial}(\lambda_i, \theta) && (22)\\
\log(\lambda_i) &= -4.14332 - 0.11911 \cdot \text{literacy}_i && (23)\\
&\quad + 0.05177 \cdot \text{ageMarried}_{15-18,i} + 0.03431 \cdot \text{ageMarried}_{18-20,i} && (24)\\
&\quad - 0.03097 \cdot \text{ageMarried}_{22-25,i} - 0.01469 \cdot \text{ageMarried}_{25-30,i} && (25)\\
&\quad - 0.00269 \cdot \text{ageMarried}_{30+,i} - 0.064757 \cdot \text{region}_{10-20k,i} && (26)\\
&\quad - 0.19794 \cdot \text{region}_{20k+,i} - 0.17982 \cdot \text{region}_{Lisbon,i} && (27)\\
&\quad + 0.09661 \cdot \text{region}_{lt10k,i} + \log(\text{monthsSinceM}_i) && (28)
\end{aligned}$$

Reference Categories:

- **Age at Marriage:** 0-15 is the reference category .
- **Region:** Porto is the reference category .

Table 3: Final Model

	Estimate	Std. Error	z value	Pr(>|z|)	Rate Ratio	2.5%	97.5%
(Intercept)	-4.143	0.064	-64.997	0.000	0.016	0.014	0.018
literacy	-0.119	0.026	-4.514	0.000	0.888	0.843	0.935
ageMarried_15_18	0.052	0.038	1.357	0.175	1.053	0.977	1.135

Table 3: Final Model

	Estimate	Std. Error	z value	Pr(>|z|)	Rate Ratio	2.5%	97.5%
ageMarried_18_20	0.034	0.031	1.096	0.273	1.035	0.973	1.100
ageMarried_22_25	-0.031	0.028	-1.094	0.274	0.970	0.917	1.025
ageMarried_25_30	-0.015	0.032	-0.458	0.647	0.985	0.925	1.049
ageMarried_30_plus	0.003	0.063	-0.043	0.966	0.997	0.882	1.128
region_10_20k	-0.065	0.067	-0.960	0.337	0.937	0.821	1.070
region_20k_plus	-0.198	0.066	-3.018	0.003	0.820	0.721	0.933
region_lisbon	-0.180	0.068	-2.639	0.008	0.835	0.731	0.955
region_lt10k	0.097	0.058	1.665	0.096	1.101	0.983	1.234

4 Conclusion

Our final model focus that both literacy and age at marriage are significant to family size across region. literacy has the most substantial impact, as it exhibits a strong negative correlation with family size, indicating that higher literacy levels are associated with smaller family sizes. Additionally, from the histogram women who marry between ages 18-20 tend to have larger families. Furthermore, less developed regions are associated with higher family sizes, suggesting that socio-economic factors play a crucial role in population.

However, age at marriage individually does not significantly affect family size, as all p-values for marriage age categories are above 0.05. Yet, when considering age as a whole, it significantly impacts family size. This suggests that variations in marriage timing alone do not drive fertility differences when controlling for literacy and regional factors. Instead, the combined effect of age at marriage, literacy, and regional differences plays a more critical role in shaping family size.

References

Conceição Rego, Conceição Freire, Isabel Joaquina Ramos, Andreia Dionísio, Maria da Saudade Baltazar, Maria Raquel Lucas. 2016. *Urban-Rural Connections and Development Perspectives in Portugal*. Center for Advanced Studies in Management; Economics – CEFAGE, University of Évora. https://esadr.uevora.pt.

Instituto Nacional de Estatística (Portugal). 1979. *Portugal World Fertility Survey (WFS), 1979-80*. World Fertility Survey (WFS), Demographic; Health Surveys (DHS) Program. https://wfs.dhsprogram.com/index.cfm?ccode=pt.

Naila Kabeer. 1996. *Gender, Demographic Transition, and the Economics of Family Size: Population Policy for a Human-Centered Development*. *UNRISD Discussion Paper*. United Nations Research Institute for Social Development (UNRISD). https://www.unrisd.org/en/library/publications/gender-demographic-transition-and-the-economics-of-family-size-population-policy-for-a-human-centered-development.

Teresa Castro Martin. 1995. *Women's Education and Fertility: Results from 26 Demographic and Health Surveys*. *Studies in Family Planning*. Vol. 26. Wiley on behalf of Population Council. https://www.jstor.org/stable/2137845.